Exercises from Ch. 4: Polynomials, Lang

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Last Updated: March 29, 2026

Disclaimer: The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

Exercises

Exercise 0.1. (Exercise 1) Let k be a field and $f(X) \in k[X]$ a non-zero polynomial. Show that the following conditions are equivalent:

a) The ideal $(f(X))$ is prime.

b) The ideal $(f(X))$ is maximal.

c) $f(X)$ is irreducible.

Proof. $(a \Rightarrow b)$ Assume $(f(X))$ is prime. Thus, by definition, $k[X]/(f(X))$ is entire. Let $\overline{g} \in k[X]/(f(X))$ be a non-zero element, i.e. $g \notin (f(X))$. Thus, $(g(X), f(X)) = 1 \Rightarrow \exists\, a(X), b(X) \in k[X]$ such that

$$a(X)g(X) + \underbrace{b(X)f(X)}_{\in (f(X))} = 1$$

Reducing this modulo $(f(X))$, we obtain $\overline{a}(X)\overline{g}(X) = \overline{1}$. Hence, $\overline{a}(X)$ is the inverse of $\overline{g}(X)$ and vice-versa. Therefore, $k[X]/(f(X))$ is a field and thus $(f(X))$ is maximal.
$(b \Rightarrow c)$ Now, assume $(f(X))$ is maximal. By way of contradiction, suppose $f(X)$ is reducible. Thus, there exists $g(X), h(X) \in k[X]$ such that $f(X) = g(X)h(X)$ where $\deg g, \deg h \geq 1$. Hence, $f(X) \in (g(X))$. Thus, $(f(X)) \subset (g(X))$. This containment is strict since $g(X) \in (g(X))$ but $g(X) \notin (f(X))$. This is because if $g(X)$ was an element in $(f(X))$, then there would exist some $r(X) \in k[X]$ such that $g(X) = f(X)r(X) \iff f(X) \Big| g(X) \iff \deg g \geq \deg f$ which would contradict the assumption that $\deg g < \deg f$ (from $f = gh$). Thus, again by our assumption, $(f(X)) \subsetneq (g(X))$. We now must show that $(g(X)) \neq k[X]$, i.e. $(g(X)) \subsetneq k[X]$. This is trivial, however, since $(g(X)) = k[X]$ if and only if $g(X)$ is a unit in $k[X]$, but the units of $k[X]$ are $(k[X])^* = k^*$, the non-zero elements of k, and $\deg g \geq 1$. Thus, there are no units of $k[X]$ in $(g(X))$. Therefore,

$$(f(X)) \subsetneq (g(X)) \subsetneq k[X]$$

contradicting the maximality of $(f(X))$ in $k[X]$. Therefore, $f(X)$ is irreducible.
$(c \Rightarrow a)$ Assume $f(X)$ is irreducible. Suppose there exists $\varphi(X), \psi(X) \in k[X]$ such that $\varphi(X)\psi(X) \in (f(X))$. Thus, there exists some $g(X) \in k[X]$ such that $\varphi(X)\psi(X) = f(X)g(X)$. Hence, $f(X) \Big| \varphi(X)\psi(X)$ (This looks a lot like Euclid's Lemma for integers). If $f(X) \Big| \varphi(X)$, then $\varphi(X) \in (f(X))$ and we are done. Thus, suppose $f(X) \nmid \varphi(X)$ (WTS $f(X) \Big| \psi(X)$). Then, $(f(X), \varphi(X)) = 1 \Rightarrow \exists\, a(X), b(X) \in k[X]$ such that $f(X)a(X) + \varphi(X)b(X) = 1$. Multiplying both sides by $\psi(X)$,

$$f(X)a(X)\psi(X) + \varphi(X)b(X)\psi(X) = \psi(X)$$

But by assumption $f(X)\Big|\varphi(X)\psi(X)$, hence, we have

$$\begin{aligned}\psi(X) &= f(X)a(X)\psi(X) + f(X)g(X)b(X) \\ &= f(X)(a(X)\psi(X) + g(X)b(X)) \Rightarrow f(X)\Big|\psi(X) \quad \because a(X)\psi(X) + g(X)b(X) \in k[X]\end{aligned}$$

Therefore, $\psi(X) \in (f(X))$. Hence, $(f(X))$ is prime.

□

Exercise 0.2. (Exercise 3)
Let f be a polynomial in one variable over a field k. Let X, Y be two variables. Show that in $k[X, Y]$ we have a "Taylor" series expansion

$$f(X+Y) = f(X) + \sum_{i=1}^{n} \varphi_i(X)Y^i,$$

where $\varphi_i(X)$ is a polynomial in X with coefficients in k. If k has characteristic 0, show that

$$\varphi_i(X) = \frac{D^i f(X)}{i!}.$$

Proof. Let $f(T) \in k[T]$. Then we can write

$$f(T) = \sum_{i=0}^{n} a_i T^i$$

with $a_i \in k$. Let $T = X + Y$. Thus,

$$\begin{aligned}f(T) = f(X+Y) &= \sum_{i=0}^{n} a_i (X+Y)^i \\ &= \sum_{i=0}^{n} a_i \sum_{k=0}^{i} \binom{i}{k} X^k Y^{i-k}.\end{aligned}$$

We wish to reorganize this based off of powers of Y. Let $j = i - k \Rightarrow k = i - j$. Hence, we have

$$f(X+Y) = \sum_{i=0}^{n} a_i \sum_{i=j}^{n} \binom{i}{i-j} X^{i-j} Y^j.$$

Fix a power j of Y. Then the corresponding coefficient of Y^j is

$$\binom{i}{i-j} X^{i-j} = \binom{i}{j} X^{i-j}.$$

Observe that we require $j = i - k \geq 0$. By our definitions, $0 \leq k \leq i$, hence $0 \leq k \Rightarrow 0 \leq i - j \Rightarrow j \leq i$ and $k \leq i \Rightarrow j \geq 0$. Thus, we restrict $0 \leq j \leq i$. Then we can write

$$\varphi_j(X) = \sum_{i=j}^{n} a_i \binom{i}{j} X^{i-j},$$

which is the coefficient of Y^j. Note that when $j = 0$, we have

$$\varphi_0(X) = \sum_{i=0}^{n} a_i X^i = f(X).$$

Thus,

$$
\begin{aligned}
f(X+Y) &= \varphi_0(X) + \sum_{j=1}^{n} \varphi_j(X) Y^j \\
&= f(X) + \sum_{j=1}^{n} \varphi_j(X) Y^j.
\end{aligned}
$$

Recall that the j-th derivative of X^i is

$$
\begin{aligned}
D^j(X^i) &= i(i-1)\cdots(i-j+1)X^{i-j} \\
&= \frac{i!}{(i-j)!j!} X^{i-j}.
\end{aligned}
$$

In our definition of $\varphi_j(X)$, observe that we can write a term of the sum representation for $\varphi_j(X)$ as

$$
\binom{i}{j} X^{i-j} = \frac{i!}{(i-j)!j!} X^{i-j} = \frac{D^j(X^i)}{j!}.
$$

Hence,

$$
\begin{aligned}
\varphi_j(X) = \sum_{i=j}^{n} a_i \binom{i}{j} X^{i-j} &= \sum_{i=j}^{n} \frac{a_i}{j!} D^j(X^i) \\
&= \frac{D^j f(X)}{j!}.
\end{aligned}
$$

□

Exercise 0.3. (Exercise 5)

a) Show that the polynomials $X^4 + 1$ and $X^6 + X^3 + 1$ are irreducible over the rational numbers.

b) Show that a polynomial of degree 3 over a field is either irreducible or has a root in the field. Is $X^3 - 5X^2 + 1$ irreducible over the rational numbers?

c) Show that the polynomial in two variables $X^2 + Y^2 - 1$ is irreducible over the rational numbers. Is it irreducible over the complex numbers?

Proof. We first state the clear facts that $\mathbb{Q}$ is the quotient field of $\mathbb{Z}$ and $\mathbb{Z}$ is a factorial ring, so Eisenstein's Criterion should be considered.

a) Let $f(X) = X^4 + 1 \in \mathbb{Q}[X]$. Consider

$$
f(X+1) = (X+1)^4 + 1 = X^4 + 4X^3 + 6X^2 + 4X + 2.
$$

Applying Eisenstein's Criterion for $p = 2$, we see that $f(X+1)$ is irreducible over $\mathbb{Q}$ and thus $f(X)$ is irreducible over $\mathbb{Q}$. Now let $g(X) = X^6 + X^3 + 1 \in \mathbb{Q}[X]$. Observe that $g(X) = \Phi_3(X^3) = \Phi_9(X) = \mathrm{Irr}(\zeta_9, \mathbb{Q}, \mathrm{X})$. One can also that $g(X)$ is irreducible by applying Eisenstein's Criterion to

$$
\begin{aligned}
g(X+1) &= (X+1)^6 + (X+1)^3 + 1 \\
&= (X^6 + 6X^5 + 15X^4 + 20X^3 + 15X^2 + 6X + 1) + (X^3 + 3X^2 + 3X + 1) + 1 \\
&= X^6 + 6X^5 + 15X^4 + 21X^3 + 18X^2 + 9X + 3
\end{aligned}
$$

with $p = 3$.

b) ????????????

c) ??????????????

c) Let D be the standard derivative in the polynomial ring $k[X]$ over a field k. Let $R(X) = c\prod(X-\alpha_i)^{m_i}$ with $\alpha_i \in K$, $c \in k$, and $m_i \in \mathbb{Z}$, so $R(X)$ is a rational function. Show that

$$R'/R = \sum \frac{m_i}{X-\alpha_i}.$$

Proof. a) Observe that since $1 = 1 \cdot 1$, we have

$$\begin{aligned} D(1) &= D(1 \cdot 1) \\ &= D(1) + D(1) = 2D(1). \end{aligned}$$

Thus, $2D(1) = D(1) \Rightarrow D(1) = 0$. Then,

$$\begin{aligned} 0 &= D(1) \\ &= D\left(\frac{x}{x}\right) \\ &= D\left(x \cdot \frac{1}{x}\right) \\ &= \frac{1}{x}Dx + xD\left(\frac{1}{x}\right) \end{aligned}$$

Subtracting and dividing gives

$$D\left(\frac{1}{x}\right) = -\frac{D(x)}{x^2}.$$

Thus for any $\alpha = x/y \in K$ $(x, y \in A)$, we have

$$\begin{aligned} D\left(\frac{x}{y}\right) &= D\left(x \cdot \frac{1}{y}\right) \\ &= \frac{1}{y}D(x) + xD\left(\frac{1}{y}\right) \\ &= \frac{yD(x) - x}{y^2} = \frac{yD(x) - xD(y)}{y^2} \qquad \because D(y) = 1 \end{aligned}$$

as desired.

b) Let $L(x) = Dx/x$ for $x \in K^\times$. Let $y \in K^\times$. Then

$$\begin{aligned} L(xy) = \frac{D(xy)}{xy} &= \frac{yD(x) + xD(y)}{xy} \\ &= \frac{D(x)}{x} + \frac{D(y)}{y} \\ &= L(x) + L(y). \end{aligned}$$

c) Let $R(X) = c\prod(X-\alpha_i)^{m_i}$ with $c, \alpha_i \in k$ and $m_i \in \mathbb{Z}$. As stated in the exercise, let D be the standard derivative in the polynomial ring $k[X]$ over k. That is,

$D\colon k[X] \to k[X]$

by $D(f(X)) = D(a_nX^n + \cdots + a_0) = na_nX^{n-1} + \cdots + a_1$.

Consider $(X-\alpha_1)^{m_1}$. Then

$$
\begin{aligned}
L\left((X-\alpha_1)^{m_1}\right) &= \frac{D\left((X-\alpha_1)^{m_1}\right)}{(X-\alpha_1)^{m_1}} \\
&= \frac{m_1(X-\alpha_1)^{m_1-1}}{(X-\alpha_1)^{m_1}} \\
&= \frac{m_1}{X-\alpha_1}.
\end{aligned}
$$

Applying the logarithmic derivative to $R(X)$, and using part (b), we obtain

$$
L(R(X)) = \frac{D(R(X))}{R(X)} = R'/R = \sum \frac{m_i}{X-\alpha_i}
$$

as desired.

□

Exercise 0.6. (Exercise 25) **Bernoulli numbers.** Define the Bernoulli numbers B_k as the coefficients in the power series

$$
F(t) = \frac{t}{e^t-1} = \sum_{k=0}^{\infty} B_k \frac{t^k}{k!}.
$$

Of course, $e^t = \sum t^n/n!$ is the standard power series with rational coefficients $1/n!$. Prove:

a) $B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$.

b) $F(-t) = t + F(t)$, and $B_k = 0$ if k is odd $\neq 1$.

Proof. a) We start with

$$
e^t = \sum_{k=0}^{\infty} \frac{t^k}{k!}.
$$

Then,

$$
e^t - 1 = \sum_{n=1}^{\infty} b_n t^n
$$

with $b_1 = 1$, $b_2 = \frac{1}{2}, \ldots, b_l = \frac{1}{l!}, \ldots$. We want a power series for $\frac{t}{e^t-1}$, i.e.,

$$
\frac{t}{e^t-1} = t \cdot \left(\frac{1}{\sum_{n=1}^{\infty} b_n t^n} \right)
$$

Observe in $e^t - 1 = t\left(1 + \frac{t}{2} + \frac{t^2}{6} + \cdots\right)$, $1 + \frac{t}{2} + \frac{t^2}{6} + \cdots \to 1$ as $t \to 0$. Thus, near 0, $e^t - 1 \sim t$. Hence

$$
\frac{1}{e^t-1} \sim \frac{1}{t}.
$$

Therefore we need a Laurent series since $\frac{1}{e^t-1}$ is not analytic at 0. Let

$$
\frac{1}{e^t-1} = \sum_{n=-1}^{\infty} c_n t^n
$$

be our Laurent series. The coefficient of t^{-1} is called the residue or pole's leading coefficient. We defined $\frac{1}{e^t-1}$ with this Laurent series since we expect a $\frac{1}{t}$ pole from our previous observation ($(e^t-1)^{-1} \sim t^{-1}$). Since our singularity is first order, not $t^{-2}, t^{-3}, \ldots$, we only need the one negative power term, i.e. $n \geq -1$. Now

$$
1 = (e^t - 1)\left(\frac{1}{e^t-1}\right) = \left(\sum_{k=1}^{\infty} b_k t^k\right)\left(\sum_{n=-1}^{\infty} c_n t^n\right)
$$

By multiplying these power series, the coefficient of t^m in the product is given by

$$\sum_{k+n=m} b_k c_n.$$

Since $\left(\sum_{k=1}^{\infty} b_k t^k\right)\left(\sum_{n=-1}^{\infty} c_n t^n\right) = 1$, the coefficient of t^0 must be 1, and the coefficient of $t^m = 0$ for $m \geq 1$. we now solve for some of the c_n. <u>($m = 0$; coeff. of t^m):</u>

$$\sum_{k+n=0} b_k c_n = b_1 c_{-1} = 1 \Rightarrow c_{-1} = 1 \quad \because b_1 = 1$$

<u>(m=1):</u>

$$\sum_{k+n=1} b_k c_n = b_1 c_0 + b_2 c_{-1} = 0 \Rightarrow c_0 = -\frac{1}{2}$$

<u>(m=2):</u>

$$\sum_{k+n=2} b_k c_n = b_1 c_1 + b_2 c_0 + b_3 c_{-1} = 0 \Rightarrow c_1 = \frac{1}{12}$$

We continue. Recall the Laurent series centered about $t = 0$ for $\frac{1}{e^t - 1}$ is

$$\sum_{n=-1}^{\infty} c_n t^n$$

Multiplying by t, we obtain the Taylor series expansion

$$\frac{t}{e^t - 1} = \sum_{n=0}^{\infty} d_n t^n = \sum_{n=0}^{\infty} B_n \frac{t^n}{n!} = F(t)$$

and therefore $d_n = c_{n-1}$. Hence,

$$\frac{t}{e^t - 1} = c_{-1} + c_0 t + c_1 t^2 + \cdots = 1 - \frac{t}{2} + \frac{t^2}{12} + \cdots$$

and by series equivalence, $d_n = \frac{B_n}{n!} \Rightarrow B_n = d_n n! = c_{n-1} n!$. Therefore, $B_0 = c_{-1} \cdot 0! = c_{-1} = 1$, $B_1 = c_0 \cdot 1! = c_0 = -\frac{1}{2}$, and $B_2 = c_1 \cdot 2! = \frac{1}{6}$, as desired.

b) Recall that we defined

$$F(t) = \frac{t}{e^t - 1} = \sum_{k=0}^{\infty} B_k \frac{t^k}{k!}.$$

Thus,

$$\begin{aligned}
F(-t) &= \frac{-t}{e^{-t} - 1} \\
&= \left(\frac{e^t}{e^t}\right)\left(\frac{-t}{e^{-t} - 1}\right) \\
&= \frac{-te^t}{1 - e^t} = \frac{t}{e^t - 1} \\
&= \frac{te^t - t + t}{e^t - 1} = \frac{t(e^t - 1) + t}{e^t - 1} \\
&= \frac{t(e^t - 1)}{e^t - 1} + \frac{t}{e^t - 1} \\
&= t + \frac{t}{e^t - 1} = t + F(t).
\end{aligned}$$

We now consider the power series of $F(-t)$ and $t + F(t)$:

$$F(-t) = \sum_{k=0}^{\infty} (-1)^k B_k \frac{t^k}{k!}$$

$$
\begin{aligned}
t + F(t) &= \sum_{k=0}^{\infty} a_k t^k + \sum_{k=0}^{\infty} B_k \frac{t^k}{k!} \\
&= \sum_{k=0}^{\infty} \left(\frac{B_k}{k!} + a_k \right) t^k
\end{aligned}
$$

with $a_1 = 1$ and $a_k = 0$ for all $k \neq 1$. Thus, by our equality,

$$
F(-t) = t + F(t) \iff \sum_{k=0}^{\infty} (-1)^k B_k \frac{t^k}{k!} = \sum_{k=0}^{\infty} \left(\frac{B_k}{k!} + a_k \right) t^k
$$

Hence,

$$
(-1)^k B_k \frac{t^k}{k!} = \frac{B_k}{k!} + a_k.
$$

If $k = 1$,

$$
-B_1 = B_1 + 1 \Rightarrow B_1 = \frac{1}{2} \quad \text{(consistent)}
$$

If $k = 2l + 1$ ($l \geq 1$, i.e. odd not equal to 1),

$$
-B_k = B_k \iff B_k = 0.
$$

Thus, if $k \neq 1$ is odd, $B_k = 0$. Note if k is even, we get $B_k = B_k$ which is consistent.

□

Exercise 0.7. (Exercise 26) **Bernoulli polynomials.** Define the Bernoulli polynomials $\mathbf{B_k}(X)$ by the power series expansion

$$
F(t, X) = \frac{te^{tX}}{e^t - 1} = \sum_{k=0}^{\infty} \mathbf{B_k}(X) \frac{t^k}{k!}.
$$

It is clear that $B_k = \mathbf{B_k}(0)$, so the Bernoulli numbers are the constant terms of the Bernoulli polynomials. Prove:

a) $\mathbf{B_0}(X) = 1$, $\mathbf{B_1}(X) = X - \frac{1}{2}$, $\mathbf{B_2}(X) = X^2 - X + \frac{1}{6}$.

b) For each positive integer N,

$$
\mathbf{B_k}(X) = N^{k-1} \sum_{a=0}^{N-1} \mathbf{B_k} \left(\frac{X + a}{N} \right).
$$

c) $\mathbf{B_k}(X) = X^k - \frac{1}{2} k X^{k-1} + \text{lower terms}$.

d) $F(t, X + 1) - F(t, X) = te^{Xt} = t \sum X^k \frac{t^k}{k!}$.

e) $\mathbf{B_k}(X + 1) - \mathbf{B_k}(X) = kX^{k-1}$ for $k \geq 1$.

Proof. a) We will utilize **Exercise 25**'s power series expansion

$$
\frac{t}{e^t - 1} = \sum_{k=0}^{\infty} d_k t^k = \sum_{k=0}^{\infty} B_k \frac{t^k}{k!},
$$

and

$$
e^{tX} = \sum_{m=0}^{\infty} \frac{(tX)^m}{m!}.
$$

Thus,

$$\begin{aligned}
\frac{te^{tX}}{e^t-1} &= \left(\frac{t}{e^t-1}\right)\left(e^{tX}\right) \\
&= \left(\sum_{k=0}^{\infty} B_k \frac{t^k}{k!}\right)\left(\sum_{m=0}^{\infty} \frac{(tX)^m}{m!}\right) \\
&= \sum_{n=0}^{\infty}\left(\sum_{k+m=n} B_k \frac{X^m}{k!m!}\right) t^n \\
&= \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} B_k \frac{X^{n-k}}{k!(n-k)!}\right) t^n \qquad \because\ m = n-k
\end{aligned}$$

From the power series expansion for $F(t,x)$, we need $\sum(\text{something})\frac{t^n}{n!}$. Thus, we multiply $\frac{1}{k!(n-k)!}$, in the inner sum, by $\frac{n!}{n!}$ which gives

$$\frac{1}{n!}\cdot\binom{n}{k}.$$

Then, our inner sum becomes

$$\sum_{k=0}^{n} B_k \frac{X^{n-k}}{k!(n-k)!} = \frac{1}{n!}\sum_{k=0}^{n}\binom{n}{k} B_k X^{n-k}$$

Substituting this back into the series,

$$\begin{aligned}
\sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} B_k \frac{X^{n-k}}{k!(n-k)!}\right) t^n &= \sum_{n=0}^{\infty}\left(\frac{1}{n!}\sum_{k=0}^{n}\binom{n}{k} B_k X^{n-k}\right) t^n \\
&= \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n}\binom{n}{k} B_k X^{n-k}\right)\frac{t^n}{n!}
\end{aligned}$$

Thus,

$$\frac{te^{tX}}{e^t-1} = \sum_{n=0}^{\infty}\mathbf{B_n}(X)\frac{t^n}{n!} = \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n}\binom{n}{k} B_k X^{n-k}\right)\frac{t^n}{n!}$$

By series equivalence, we obtain,

$$\mathbf{B_n}(X) = \sum_{k=0}^{n}\binom{n}{k} B_k X^{n-k}$$

Therefore,

- $\mathbf{B_0}(X) = \binom{n}{k} B_0 X^0 = B_0 = 1$
- $\mathbf{B_1}(X) = \binom{1}{0} B_0 X^1 + \binom{1}{1} B_1 X^0 = B_0 X + B_1 = X - \frac{1}{2}$
- $\mathbf{B_2}(X) = \binom{2}{0} B_0 X^2 + \binom{2}{1} B_1 X^1 + \binom{2}{2} B_2 X^0 = B_0 X^2 + 2B_1 X + B_2 = X^2 - X + \frac{1}{6}$

b) ??

c) ??

d) ??

e) ??

□